COURSE 2 · ALGEBRA

Algebra

Printable workbook with worked solutions

64 problems across 8 modules, plus the course capstone.
Full worked solutions at the back.

Name _____ Started _____

Before you start

How to use this workbook

There are 64 problems, eight per module, followed by the course capstone. Dots show difficulty: one dot is a warm-up, two is the standard you are aiming for, three stretches towards first-year university.

Work in pencil and show your steps. Try each problem before looking at the solutions, which start on the page headed *Worked solutions*. The same problems are in the interactive workbook, which checks answers instantly and adds a quick quiz for each module.

> **A good routine for each module**
>
> Problems 1–2 straight after listening to the reader. Problems 3–6 after the explorer. Problems 7–8 at the end of the week, then redo anything you got wrong without looking.

Module 2.1

Expressions and simplifying

2.1.1 ●○○

Simplify $4x + 3y - x + 2y$.

A. $3x + 5y$

B. $5x + 5y$

C. $8xy$

D. $3x + y$

2.1.2 ●○○

Expand $5(2x - 3)$.

A. $10x - 3$

B. $10x - 15$

C. $7x - 8$

D. $10x + 15$

2.1.3 ●●○

Expand $(x + 4)(x - 2)$.

A. $x^2 + 2x - 8$

B. $x^2 - 8$

C. $x^2 + 6x - 8$

D. $x^2 - 2x + 8$

2.1.4 ●●○

Factorise $x^2 - 7x + 12$.

A. $(x - 3)(x - 4)$

B. $(x + 3)(x + 4)$

C. $(x - 2)(x - 6)$

D. $(x - 12)(x + 1)$

2.1.5 ●●○

Factorise $9x^2 - 25$.

A. $(3x - 5)^2$

B. $(3x - 5)(3x + 5)$

C. $(9x - 25)(x + 1)$

D. Cannot be factorised

2.1.6 ●●○

Evaluate 8^(2/3).

Answer: _____

2.1.7 ●●○

Simplify $(x^3 \times x^5) \div x^2$.

A. x^6

B. x^8

C. $x^{15}/^2$

D. x^{10}

2.1.8 ●●●

If $x = 3$ and $y = -2$, evaluate $2x^2 - 3xy + y^2$.

Answer: _____

Module 2.2

Linear equations and inequalities

2.2.1 ●○○

Solve $4x + 7 = 31$.

Answer: _____

2.2.2 ●○○

Solve $3(x - 2) = 15$.

Answer: _____

2.2.3 ●●○

Solve $x/4 + 2 = x/2 - 1$.

Answer: _____

2.2.4 ●●○

Solve $5 - 2x > 11$.

A. $x > -3$

B. $x < -3$

C. $x > 3$

D. $x < 3$

2.2.5 ●●○

Find the gradient of the line through $(1, 3)$ and $(5, 11)$.

Answer: _____

2.2.6 ●●○

A taxi charges a $4 flag fall plus $2.50 per km. How far can you go for $29?

Answer: _____ km

2.2.7 ●●○

On a graph of distance (m) against time (s), the gradient represents

A. acceleration

B. speed

C. distance

D. time

2.2.8 ●●●

Temperature in °F is $F = 1.8C + 32$. At what temperature are Celsius and Fahrenheit equal?

Answer: _____ °

Module 2.3

Rearranging formulas

2.3.1 ●○○

Final velocity: v = u + at.

$$v = u + at$$

Make **a** the subject.

2.3.2 ●○○

Ideal gas law: PV = nRT.

$$PV = nRT$$

Make **T** the subject.

2.3.3 ●●○

Kinetic energy: E = ½mv². (Type the half as 1/2 if needed.)

$$E = \tfrac{1}{2}mv^2$$

Make **v** the subject.

2.3.4 ●●○

Pendulum: T = 2π√(L/g).

$$T = 2\pi\sqrt{\tfrac{L}{g}}$$

Make **L** the subject.

2.3.5 ●●○

Lens equation: 1/f = 1/u + 1/v.

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Make **v** the subject.

2.3.6 ●●●

Solve for x: y = (x + 2)/(x − 3).

$$y = \frac{x+2}{x-3}$$

Make **x** the subject.

2.3.7 ●●○

A pendulum clock must have a period of 2.00 s where g = 9.81 m/s². How long is the pendulum?

Answer: _____ m

2.3.8 ●●○

From 1/f = 1/u + 1/v, a student writes f = u + v. The mistake is

A. a sign error

B. flipping individual terms

C. forgetting to square

D. nothing

Module 2.4

Functions and graphs

2.4.1 ●○○

If $f(x) = 3x^2 - 2$, find $f(4)$.

Answer: _____

2.4.2 ●○○

What is the domain of $f(x) = \sqrt{(x - 5)}$?

A. all real x

B. $x \geq 5$

C. $x \geq 0$

D. $x > 5$ only

2.4.3 ●●○

The graph of $y = (x + 3)^2$ is the graph of $y = x^2$ shifted

A. 3 right

B. 3 left

C. 3 up

D. 3 down

2.4.4 ●●○

The vertex of $y = 2(x - 1)^2 + 5$ is at

A. (1, 5)

B. (−1, 5)

C. (2, 5)

D. (1, −5)

2.4.5 ●●○

If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(3))$.

Answer: _____

2.4.6 ●●○

What is the inverse of $f(x) = (x - 4)/3$?

A. $3x + 4$

B. $3x - 4$

C. $(x + 4)/3$

D. $1/(3x - 4)$

2.4.7 ●●○

Which function has a vertical asymptote at $x = 2$?

A. $y = x - 2$

B. $y = 1/(x - 2)$

C. $y = (x - 2)^2$

D. $y = 2^x$

2.4.8 ●●●

Sketch in words how $y = -2(x - 3)^2 + 4$ relates to $y = x^2$.

Module 2.5

Systems of equations

2.5.1 ●○○

Solve x + y = 10 and x − y = 4.

A. x = 7, y = 3

B. x = 6, y = 4

C. x = 3, y = 7

D. x = 5, y = 5

2.5.2 ●●○

Solve 3x + 2y = 16 and x − y = 2. What is x?

Answer: _____

2.5.3 ●●○

Solve 2x + 5y = 19 and 3x − 2y = 0. What is y?

Answer: _____

2.5.4 ●●○

Adult tickets cost $15 and child tickets $8. 40 tickets raised $467. How many adult tickets were sold?

Answer: _____

2.5.5 ●●○

The system $y = 2x + 1$ and $y = 2x - 3$ has

A. one solution

B. no solution

C. infinitely many

D. two solutions

2.5.6 ●●○

How many litres of 10% salt solution must be mixed with 30% solution to make 20 L of 25% solution?

Answer: _____ L

2.5.7 ●●●

Solve $x^2 + y = 7$ and $y = x + 1$ for the positive value of x.

Answer: _____

2.5.8 ●●●

Explain geometrically why two linear equations in two unknowns can have exactly one, none, or infinitely many solutions, but never exactly two.

Module 2.6

Quadratics and polynomials

2.6.1 ●○○

Solve $x^2 - 9x + 20 = 0$.

A. 4 and 5

B. −4 and −5

C. 2 and 10

D. 1 and 20

2.6.2 ●●○

Use the quadratic formula to find the larger root of $2x^2 - 3x - 5 = 0$.

Answer: _____

2.6.3 ●●○

Find the discriminant of $3x^2 + 4x + 2$.

Answer: _____

2.6.4 ●●○

Find the x-coordinate of the vertex of $y = x^2 - 6x + 11$.

Answer: _____

2.6.5 ●●○

A ball's height is $h = 20t - 5t^2$ metres. After how many seconds does it land?

Answer: _____ s

2.6.6 ●●○

For the same ball, $h = 20t - 5t^2$, what is the maximum height?

Answer: _____ m

2.6.7 ●●○

$p(x) = x^3 - 6x^2 + 11x - 6$ and $p(1) = 0$. Which is a factor?

A. $x + 1$

B. $x - 1$

C. $x - 6$

D. $x + 6$

2.6.8 ●●●

For which value of k does $x^2 + kx + 9 = 0$ have exactly one root ($k > 0$)?

A. 3

B. 6

C. 9

D. 18

Module 2.7

Exponentials and logarithms

2.7.1 ●○○

Evaluate $\log_{10} 10\,000$.

Answer: _____

2.7.2 ●○○

A sample has a half-life of 6 hours. What fraction remains after 18 hours? Give a decimal.

Answer: _____

2.7.3 ●●○

Solve $2^x = 50$, to 3 significant figures.

Answer: _____

2.7.4 ●●○

$1000 is invested at 5% a year compounded annually. What is it worth after 10 years?

Answer: _____ dollars

2.7.5 ●●○

Using the rule of 70, estimate the doubling time for growth of 7% a year.

Answer: _____ years

2.7.6 ●●○

A substance decays with k = 0.1 per day. What is its half-life?

Answer: _____ days

2.7.7 ●●○

How many times more acidic is a solution of pH 2 than one of pH 5?

Answer: _____ ×

2.7.8 ●●●

Simplify $\ln(e^5) + \log_{10}(100)$.

A. 7

B. 5

C. 10

D. 500

Module 2.8

Calculus intuition

2.8.1 ●○○

Differentiate $y = x^4$.

A. $4x^3$

B. x^3

C. $4x^4$

D. $x^5/5$

2.8.2 ●○○

Differentiate $y = 3x^2 + 5x - 7$.

A. $6x + 5$

B. $3x + 5$

C. $6x - 7$

D. $6x^2 + 5$

2.8.3 ●●○

For $y = x^3 - 2x$, find the gradient at $x = 2$.

Answer: _____

2.8.4 ●●○

A ball's height is $h = 30t - 5t^2$. What is its velocity, dh/dt, at $t = 1$ s?

Answer: _____ m/s

2.8.5 ●●○

For $h = 30t - 5t^2$, at what time is the ball at its highest?

Answer: _____ s

2.8.6 ●●○

A car's speed rises steadily from 0 to 20 m/s in 10 s. How far does it travel in that time (area under the graph)?

Answer: _____ m

2.8.7 ●●○

Evaluate the integral of $2x$ from $x = 0$ to $x = 3$.

Answer: _____

2.8.8 ●●●

Explain in words what $dV/dt = -0.5$ litres per minute means for a draining tank.

Capstone

Modelling a cooling cup of coffee

A cup of coffee at 90 °C is left in a 20 °C room. Newton's law of cooling says the temperature difference from the room decays exponentially: T = 20 + 70e^(−kt), with t in minutes. After 10 minutes the coffee is at 60 °C. Use algebra from across the course to build and use the model.

Part 1 ●●○

Write 60 = 20 + 70e^(−10k) and solve for k (per minute).

Answer: _____ per min

Part 2 ●●○

What is the coffee's temperature after 30 minutes?

Answer: _____ °C

Part 3 ●●○

You like coffee at 50 °C. Rearrange to find when it reaches 50 °C (minutes).

Answer: _____ min

Part 4 ●●●

Differentiate to find the rate of cooling at t = 0 (degrees per minute, give the size only).

Answer: _____ °C/min

Part 5 ●●○

To check the model against measurements, which graph should be a straight line?

A. T against t

B. ln(T − 20) against t

C. T against ln t

D. 1/T against t

Part 6 ●●●

Name one assumption of the model and one situation where it would fail.

Answers

Worked solutions

Module 2.1 · Expressions and simplifying

2.1.1 ANSWER A. $3x + 5y$

1 Collect x terms: $4x - x = 3x$.

2 Collect y terms: $3y + 2y = 5y$.

2.1.2 ANSWER B. $10x - 15$

1 Multiply each term by 5.

2.1.3 ANSWER A. $x^2 + 2x - 8$

1 $x \cdot x + x \cdot (-2) + 4 \cdot x + 4 \cdot (-2)$.

2 $x^2 + 2x - 8$.

2.1.4 ANSWER A. $(x - 3)(x - 4)$

1 Two numbers with sum -7 and product 12: -3 and -4.

2.1.5 ANSWER B. $(3x - 5)(3x + 5)$

1 Difference of two squares: $(3x)^2 - 5^2$.

2.1.6 ANSWER 4

1 Cube root of 8 is 2.

2 $2^2 = 4$.

2.1.7 ANSWER A. x^6

1 Multiply: add exponents to get x^8.

2 Divide: subtract to get x^6.

2.1.8 ANSWER 40

1 $2(9) - 3(3)(-2) + (-2)^2$.

2 $18 + 18 + 4 = 40$.

Module 2.2 · Linear equations and inequalities

2.2.1 ANSWER 6

1 Subtract 7: $4x = 24$.

2 Divide by 4: $x = 6$.

2.2.2 ANSWER 7

1 Expand: 3x − 6 = 15.

2 3x = 21, x = 7.

2.2.3 ANSWER 12

1 Multiply everything by 4: x + 8 = 2x − 4.

2 12 = x.

2.2.4 ANSWER B. x < −3

1 −2x > 6.

2 Divide by −2 and flip: x < −3.

2.2.5 ANSWER 2

1 (11 − 3) ÷ (5 − 1) = 8 ÷ 4.

2 = 2.

2.2.6 ANSWER 10 km

1 29 = 4 + 2.5d.

2 2.5d = 25, d = 10 km.

2.2.7 ANSWER B. speed

1 Gradient = change in distance ÷ change in time = speed.

2.2.8 ANSWER -40 °

1 Set F = C: C = 1.8C + 32.

2 −0.8C = 32, C = −40.

Module 2.3 · Rearranging formulas

2.3.1 ANSWER a = (v − u)/t

1 Subtract u from both sides.

2 Divide by t.

2.3.2 ANSWER T = PV/(nR)

1 Divide both sides by nR.

2.3.3 ANSWER v = √(2E/m)

1 Multiply by 2: $2E = mv^2$.

2 Divide by m, then take the positive square root.

2.3.4 ANSWER $L = gT^2/(4\pi^2)$

1 Divide by 2π.

2 Square both sides: $T^2/4\pi^2 = L/g$.

3 Multiply by g.

2.3.5 ANSWER $v = uf/(u - f)$

1 Subtract $1/u$: $1/v = 1/f - 1/u$.

2 Combine: $(u - f)/(uf)$.

3 Flip both sides.

2.3.6 ANSWER $x = (3y + 2)/(y - 1)$

1 Multiply by $(x - 3)$ and expand: $xy - 3y = x + 2$.

2 Collect: $xy - x = 3y + 2$.

3 Factor and divide: $x = (3y + 2)/(y - 1)$.

2.3.7 ANSWER 0.994 m

1 $L = gT^2/(4\pi^2)$.

2 $= 9.81 \times 4 \div 39.48 \approx 0.994$ m.

2.3.8 ANSWER B. flipping individual terms

1 The reciprocal of a sum is not the sum of reciprocals: $1/2 + 1/2 = 1$, but $2 + 2 \neq 1$.

Module 2.4 · Functions and graphs

2.4.1 ANSWER 46

1 $3 \times 16 - 2$.

2 $= 46$.

2.4.2 ANSWER B. $x \geq 5$

1 The quantity under the root must be ≥ 0: $x - 5 \geq 0$.

2.4.3 ANSWER B. 3 left

1 $x - h$ with $h = -3$: shift left by 3.

2.4.4 ANSWER A. (1, 5)

1 Vertex form $a(x - h)^2 + k$ has vertex (h, k).

2.4.5 ANSWER 19

1 $g(3) = 9$.

2 $f(9) = 19$.

2.4.6 ANSWER A. $3x + 4$

1 Write $y = (x - 4)/3$ and solve for x: $x = 3y + 4$.

2 So $f^{-1}(x) = 3x + 4$.

2.4.7 ANSWER B. $y = 1/(x - 2)$

1 Division by zero at $x = 2$.

2.4.8 ANSWER Shift right 3, stretch vertically by a factor of 2, reflect upside down, then shift up 4. It is an upside-down parabola, narrower than $y = x^2$, with its highest point at $(3, 4)$.

1 Read off h, k, and the sign and size of a.

Module 2.5 · Systems of equations

2.5.1 ANSWER A. $x = 7, y = 3$

1 Add: $2x = 14$, $x = 7$.

2 $y = 3$.

2.5.2 ANSWER 4

1 From the second, $x = y + 2$.

2 $3(y + 2) + 2y = 16 \rightarrow 5y = 10 \rightarrow y = 2, x = 4$.

2.5.3 ANSWER 3

1 Multiply first by 3, second by 2: $6x + 15y = 57$, $6x - 4y = 0$.

2 Subtract: $19y = 57$, $y = 3$.

2.5.4 ANSWER 21

1 $a + c = 40$, $15a + 8c = 467$.

2 $15a + 8(40 - a) = 467 \rightarrow 7a = 147 \rightarrow a = 21$.

2.5.5 ANSWER B. no solution

1 Same gradient, different intercepts: parallel lines.

2.5.6 ANSWER 5 L

1 $x + y = 20$; $0.1x + 0.3y = 5$.

2 $0.1x + 0.3(20 - x) = 5 \rightarrow -0.2x = -1 \rightarrow x = 5$ L.

2.5.7 ANSWER 2

1 Substitute: $x^2 + x + 1 = 7 \to x^2 + x - 6 = 0$.

2 $(x + 3)(x - 2) = 0$, positive root $x = 2$.

2.5.8 ANSWER Each equation is a straight line. Two distinct straight lines either cross at exactly one point or are parallel and never meet. If they are the same line, every point is shared. Two straight lines cannot cross twice.

1 What can two straight lines do?

Module 2.6 · Quadratics and polynomials

2.6.1 ANSWER A. 4 and 5

1 Factorise: $(x - 4)(x - 5) = 0$.

2.6.2 ANSWER 2.5

1 $\Delta = 9 + 40 = 49$.

2 $x = (3 \pm 7)/4$: 2.5 or −1.

2.6.3 ANSWER -8

1 $b^2 - 4ac = 16 - 24$.

2 $= -8$, so no real roots.

2.6.4 ANSWER 3

1 $x = -b/2a = 6/2$.

2 $= 3$.

2.6.5 ANSWER 4 s

1 $h = 0$: $5t(4 - t) = 0$.

2 $t = 0$ (launch) or $t = 4$ s.

2.6.6 ANSWER 20 m

1 Vertex at $t = -20/(2 \times -5) = 2$ s.

2 $h = 40 - 20 = 20$ m.

2.6.7 ANSWER B. x − 1

1 Factor theorem: $p(1) = 0$ means $(x - 1)$ is a factor.

2.6.8 ANSWER B. 6

1 $\Delta = k^2 - 36 = 0$.

2 $k = 6$.

Module 2.7 · Exponentials and logarithms

2.7.1 ANSWER 4

1 $10^4 = 10\,000$.

2.7.2 ANSWER 0.125

1 18 h = 3 half-lives.

2 $(½)^3 = 0.125$.

2.7.3 ANSWER 5.644

1 x = ln 50 ÷ ln 2.

2 = 3.912 ÷ 0.693 ≈ 5.64.

2.7.4 ANSWER 1629 dollars

1 1000×1.05^{10}.

2 ≈ $1629.

2.7.5 ANSWER 10 years

1 70 ÷ 7 = 10 years.

2.7.6 ANSWER 6.931 days

1 $t½ = \ln 2 / k$.

2 0.693 ÷ 0.1 ≈ 6.93 days.

2.7.7 ANSWER 1000 ×

1 Each pH unit is a factor of 10.

2 $10^3 = 1000$.

2.7.8 ANSWER A. 7

1 $\ln e^5 = 5$; log 100 = 2.

2 5 + 2 = 7.

Module 2.8 · Calculus intuition

2.8.1 ANSWER A. $4x^3$

1 Bring down 4, reduce power to 3.

2.8.2 ANSWER A. 6x + 5

1 3 × 2x + 5 + 0.

2.8.3 ANSWER 10

1 $dy/dx = 3x^2 - 2$.

2 At x = 2: 12 − 2 = 10.

2.8.4 ANSWER 20 m/s

1 $dh/dt = 30 - 10t$.

2 At t = 1: 20 m/s.

2.8.5 ANSWER 3 s

1 Set $dh/dt = 0$: $30 - 10t = 0$.

2 t = 3 s.

2.8.6 ANSWER 100 m

1 Triangle area = ½ × base × height.

2 ½ × 10 × 20 = 100 m.

2.8.7 ANSWER 9

1 Antiderivative: x^2.

2 $3^2 - 0^2 = 9$.

2.8.8 ANSWER The volume of water V is changing over time t at a rate of −0.5 litres per minute: the tank is losing half a litre every minute at that moment. The negative sign means volume is decreasing.

1 Read dV/dt as 'rate of change of V with time'.

Capstone · Modelling a cooling cup of coffee

Part 1 ANSWER 0.05596 per min

1 Subtract 20: 40 = 70e^(−10k).

2 Divide: e^(−10k) = 4/7. Take ln: −10k = ln(4/7).

3 k = ln(7/4)/10 ≈ 0.056 per min.

Part 2 ANSWER 33.06 °C

1 $e^{(-30k)} = (4/7)^3 \approx 0.187$.

2 T = 20 + 70 × 0.187 ≈ 33 °C.

Part 3 ANSWER 15.14 min

1 30 = 70e^(−kt) → t = ln(70/30)/k.

2 = 0.847 ÷ 0.0560 ≈ 15.1 min.

Part 4 ANSWER 3.917 °C/min

1 dT/dt = −70k e^(−kt).

2 At t = 0: −70k ≈ −3.9 °C per minute.

Part 5 ANSWER B. ln(T − 20) against t

1 T − 20 = 70e^(−kt), so ln(T − 20) = ln 70 − kt: straight line, gradient −k.

Part 6 ANSWER It assumes the room stays at 20 °C and the rate of heat loss is proportional to the temperature difference. It would fail if the room warmed up, if the cup had a lid added part-way, or if evaporation from very hot coffee made early cooling faster than predicted.

1 What must stay constant for the exponential to hold?